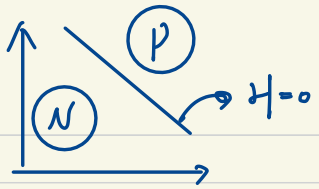
(線性)

LDA :



想找超平面H → D(X) 我們的 decision boundary.

$H \in \mathbb{R}^d$ 是一個d維超平面

$\Rightarrow w_1x_1 + w_2x_2 + \dots + x_d w_d + b = 0$

↳ bias (超平面偏差值)

$\Rightarrow [w_1\ w_2 \cdots w_d] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} + b = 0$ (H的函式)

我們令 $w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \in \mathbb{R}^d \rightarrow D(x) = w^T x + b = 0$

How to find w? b?

$w^T = (\mu_1 - \mu_2)^T \Sigma^{-1}$ $\quad b = -\frac{1}{2}(\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 + \mu_2) - \ln\left(\frac{C_1 P_2}{C_2 P_1}\right)$

μ_1, μ_2 : 正負類別訓練資料之平均向量 (1段設各N筆資料)

$$\vec{\mu} = \left.\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}_{n=1} + \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}_{2} \dots + \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}_{N} \right/ N = \begin{bmatrix} \sum_{n=1}^{N} x_1 / N \\ \sum_{n=1}^{N} x_2 / N \\ \vdots \\ \sum_{n=1}^{N} x_d / N \end{bmatrix} \qquad \left(\mu = \frac{1}{N}\sum_{n=1}^{N} x_n\right)$$

$\Sigma \in \mathbb{R}^{d \times d}$ 為正負類別訓練資料的共變數矩陣

$$\begin{cases} \Sigma_1 = \frac{1}{N_1 - 1} \sum_{n=1}^{N_1} (x_n - \mu_1)(x_n - \mu_1)^T \\ \Sigma_2 = \frac{1}{N_2 - 1} \sum_{j=1}^{N_2} (x_j - \mu_2)(x_j - \mu_2)^T \end{cases}$$

$\Rightarrow \Sigma = P_1 \Sigma_1 + P_2 \Sigma_2$

P_1 ↓ 正樣本發生率 $\quad P_2$ ↓ 負樣本發生率

Ex. Positive class: $\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \end{bmatrix} \end{array}$, Negative class: $\begin{array}{c} \begin{array}{ccc} x_4 & x_5 & x_6 \end{array} \\ \begin{bmatrix} -4 & -3 & -2 \\ -7 & -2 & 0 \end{bmatrix} \end{array}$

$$\mu_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} x_i = \frac{1}{3}\left(\begin{bmatrix}1\\2\end{bmatrix}+\begin{bmatrix}2\\5\end{bmatrix}+\begin{bmatrix}3\\2\end{bmatrix}\right) = \frac{1}{3}\left(\begin{bmatrix}6\\9\end{bmatrix}\right) = \begin{bmatrix}2\\3\end{bmatrix}$$

$$\mu_2 = \frac{1}{n_2}\sum_{j=1}^{n_2} x_j = \frac{1}{3}\left(\begin{bmatrix}-4\\-7\end{bmatrix}+\begin{bmatrix}-3\\-2\end{bmatrix}+\begin{bmatrix}-2\\0\end{bmatrix}\right) = \frac{1}{3}\left(\begin{bmatrix}-9\\-9\end{bmatrix}\right) = \begin{bmatrix}-3\\-3\end{bmatrix}$$

$$\Sigma_1 = \frac{1}{n_1-1}\sum_{i=1}^{n_1}(x_i-\mu_1)(x_i-\mu_1)^T$$

$$= \frac{1}{2}\left\{\left(\begin{bmatrix}1-2\\2-3\end{bmatrix}\begin{bmatrix}1-2 & 2-3\end{bmatrix}\right)+\left(\begin{bmatrix}2-2\\5-3\end{bmatrix}\begin{bmatrix}2-2 & 5-3\end{bmatrix}\right)+\left(\begin{bmatrix}3-2\\2-3\end{bmatrix}\begin{bmatrix}3-2 & 2-3\end{bmatrix}\right)\right\}$$

$$= \frac{1}{2}\left\{\left(\begin{bmatrix}-1\\-1\end{bmatrix}\begin{bmatrix}-1 & -1\end{bmatrix}\right)+\left(\begin{bmatrix}0\\2\end{bmatrix}\begin{bmatrix}0 & 2\end{bmatrix}\right)+\left(\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix}\right)\right\}$$

$$= \frac{1}{2}\left\{\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}+\begin{bmatrix}0 & 0\\0 & 4\end{bmatrix}+\begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix}\right\}$$

$$= \frac{1}{2}\left\{\begin{bmatrix}2 & 0\\0 & 6\end{bmatrix}\right\}$$

$$= \begin{bmatrix}1 & 0\\0 & 3\end{bmatrix}$$

$$\Sigma_2 = \frac{1}{n_2-1}\sum_{j=1}^{n_2}(x_j-\mu_2)(x_j-\mu_2)^T$$

$$= \frac{1}{2}\left\{\begin{bmatrix}-1\\-4\end{bmatrix}\begin{bmatrix}-1 & -4\end{bmatrix}+\begin{bmatrix}0\\1\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}+\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & 3\end{bmatrix}\right\}$$

$$= \frac{1}{2}\left\{\begin{bmatrix}1 & 4\\4 & 16\end{bmatrix}+\begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}+\begin{bmatrix}1 & 3\\3 & 9\end{bmatrix}\right\}$$

$$= \frac{1}{2}\begin{bmatrix}2 & 7\\7 & 26\end{bmatrix}$$

$$= \begin{bmatrix}1 & 3.5\\3.5 & 13\end{bmatrix}$$

$$\Sigma = P_1\Sigma_1 + P_2\Sigma_2 = \frac{3}{6}\begin{bmatrix}1 & 0\\0 & 3\end{bmatrix}+\frac{3}{6}\begin{bmatrix}1 & 3.5\\3.5 & 13\end{bmatrix} = \begin{bmatrix}1 & 1.75\\1.75 & 8\end{bmatrix} \#$$

求 w^T, b？已知 $\mu_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \mu_2 = \begin{bmatrix} -3 \\ -3 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 1.75 \\ 1.75 & 8 \end{bmatrix}$

$$w^T = (\mu_1 - \mu_2)^T \Sigma^{-1}$$

$$= \begin{bmatrix} 2-(-3) \\ 3-(-3) \end{bmatrix}^T \begin{bmatrix} -1.6203 & -0.3544 \\ -0.3544 & 0.2025 \end{bmatrix}$$

$$= [5 \ \ 6] \Sigma^{-1}$$

$$= [5.9746 \ \ -0.556]$$

$$\Sigma^{-1} = \frac{1}{4.9375} \begin{bmatrix} 8 & -1.75 \\ -1.75 & 1 \end{bmatrix} = \begin{bmatrix} 1.6203 & -0.3544 \\ -0.3544 & 0.2025 \end{bmatrix}$$

C_1 正類別懲罰權重

C_2 負類別懲罰權重

令 $C_1 = C_2 = 1$

$$\ln\left(\frac{0.5}{0.5}\right) = \ln(1) = 0$$

$$b = -\frac{1}{2}(\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 + \mu_2) - \ln\left(\frac{C_1 P_2}{C_2 P_1}\right)$$

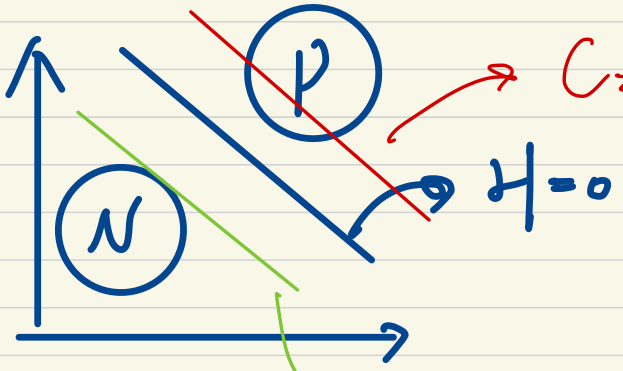
$$= -\frac{1}{2}[5 \ \ 6] \Sigma^{-1} \cancel{\begin{bmatrix} 5 \\ 6 \end{bmatrix}} - \underbrace{\ln\left(\frac{C_1 \frac{1}{2}}{C_2 \frac{1}{2}}\right)}_{\to 0}$$

$$= -\frac{1}{2}[5 \ \ 6] \begin{bmatrix} -1.62 & -0.35 \\ -0.35 & 0.2 \end{bmatrix} \cancel{\begin{bmatrix} 5 \\ 6 \end{bmatrix}}$$

$$= -\frac{1}{2}[-10.2 \ \ -0.55] \cancel{\begin{bmatrix} 5 \\ 6 \end{bmatrix}}$$

$$= -27.16$$

$$\boxed{\begin{aligned} C_1 P_2 &= C_2 P_1 \\ C_1 : C_2 &= P_1 : P_2 \end{aligned}}$$



$C_2 > C_1$ 造成更多 FP

$C_1 > C_2$ 造成更多 FN

QDA.

$qdf(x)$

$$= \frac{1}{2}x^T(\Sigma_1^{-1} - \Sigma_2^{-1})x + (\mu_1^T\Sigma_1^{-1} - \mu_2^T\Sigma_2^{-1})x - \frac{1}{2}\ln\left(\frac{\Sigma_1}{\Sigma_2}\right)$$

$$- \frac{1}{2}(\mu_1^T\Sigma_1^{-1}\mu_1 - \mu_2^T\Sigma_2^{-1}\mu_2) - \ln\left(\frac{C_{12}}{C_{21}}\frac{\pi_2}{\pi_1}\right)$$

MDA

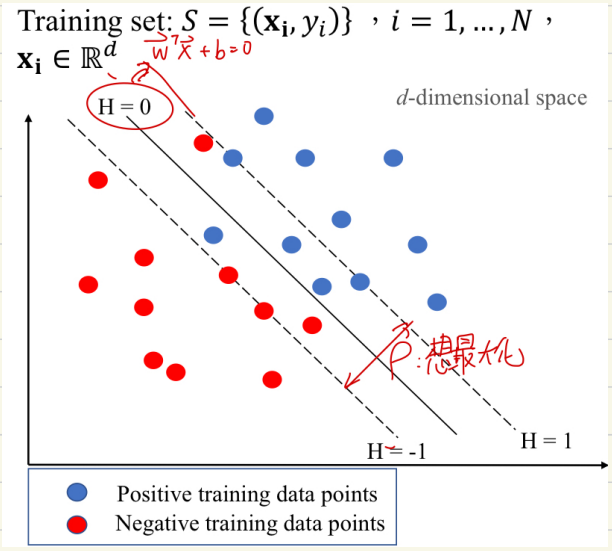
$mdf(x)$

$$= -x^T(\Sigma_1^{-1} - \Sigma_2^{-1})x + 2(\mu_1^T\Sigma_1^{-1} - \mu_2^T\Sigma_2^{-1})x - \mu_1^T\Sigma_1^{-1}\mu_1 - \mu_2^T\Sigma_2^{-1}\mu_2$$

LDA	QDA	MDA
線性鑑別分析LDA分類器假設類別的資料具有常態分佈(Normal distribution)與相同的共變異矩陣(Covariance matrix)	二次鑑別分析QDA分類器不同於線性鑑別分析分類器在於類別的資料不需要有相同的共變異矩陣	馬氏鑑別分析MDA分類器與二次鑑別分析分類器很相似，不過還是有些例外。第一點，馬氏鑑別分析分類器假設所有類別的先前機率完全相同，第二點，此分類基於馬氏距離平方。

假設分佈一樣，只是平均值不一樣

SVD 支持向量機



1° maximize ρ 分離邊界

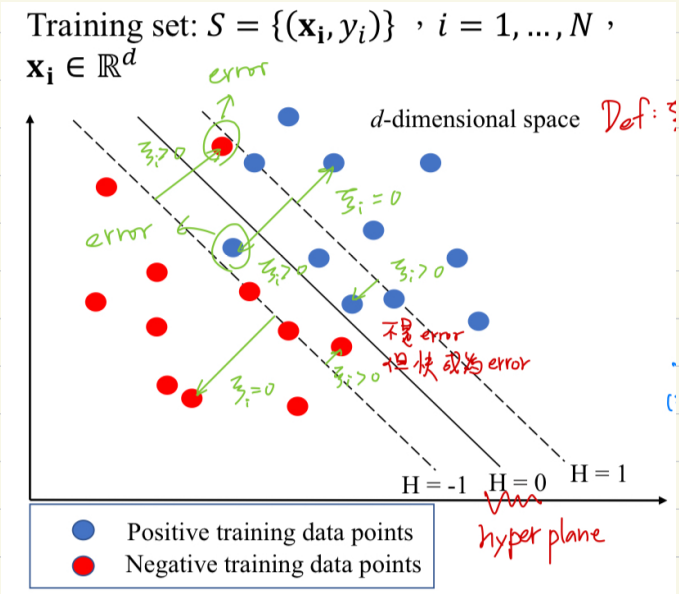
$$\rho = \frac{2}{\|\vec{w}\|}$$ 權重

$$\max \frac{2}{\|\vec{w}\|}$$

$$\Leftrightarrow \min \frac{\|\vec{w}\|}{2}$$

$$\Leftrightarrow \min \frac{\|\vec{w}\|^2}{2}$$

$$\Leftrightarrow \min \frac{1}{2} \vec{w}^T \vec{w}$$



2° minimize training error ξ_i

Def ξ:

For $\vec{x}_i \in$ positive class

If $\vec{x}_i \Rightarrow \vec{w}^T\vec{x}_i + b \geq 1$, $\xi_i = 0$

else $\xi_i > 0$

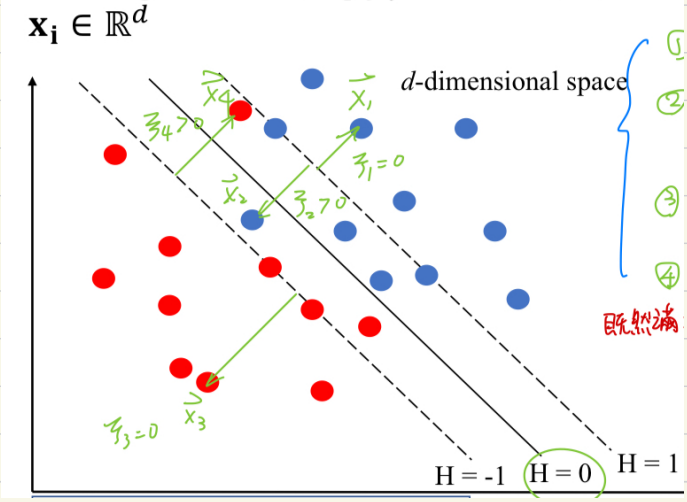
For $\vec{x}_i \in$ negative class

If $\vec{x}_i \Rightarrow \vec{w}\vec{x}_i + b \leq -1$, $\xi_i = 0$

else $\xi_i > 0$

$$\min \sum^{N} \xi_i = \xi_1 + \xi_2 + \ldots, \xi_N$$

Training set: $S = \{(\mathbf{x_i}, y_i)\}$, $i = 1, \dots, N$,
$\mathbf{x_i} \in \mathbb{R}^d$



Case 1° $y=1, \xi_1=0$

$\vec{w}^T\vec{x}_1+b \geq 1-\xi_1$

$\Rightarrow y_1(\vec{w}^T\vec{x}_1+b) \geq 1-\xi_1$

Case 2° $y=1, \xi_2>0$

$\vec{w}^T\vec{x}_2+b = 1-\xi_2$

$\Rightarrow \vec{w}^T\vec{x}_2+b \geq 1-\xi_2$

$\Rightarrow y_2(\vec{w}^T\vec{x}_2+b) \geq 1-\xi_2$

Case 3° $y=-1, \xi_3=0$

$\vec{w}^T\vec{x}_3+b \leq -1+\xi_3$

$y_3(\vec{w}^T\vec{x}_3+b) \geq 1-\xi_3$

Case 4° $y=1, \xi_4=1$

$\vec{w}^T\vec{x}_4+b = -1+\xi_4$

$\vec{w}^T\vec{x}_4+b \leq -1+\xi_4$

$y_4(w^T\vec{x}_4+b) \geq 1-\xi_4$

Summary: $y_i(\vec{w}^T\vec{x}_i+b) \geq 1-\xi_i$

$\Rightarrow y_i(\vec{w}^T\vec{x}_i+b)-1+\xi_i \geq 0 \; ; \; \xi_i \geq 0, \forall i$

Problem Formulation

ξ 鬆弛變數

$$\min_{(w,b)} \frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{N}\xi_i$$

C是懲罰權重

s.t. $y_i(\vec{w}^T\vec{x}_i+b)-1+\xi_i \geq 0, \forall i=1,\dots,N$

$\xi_i \geq 0, \forall i=1,\dots,N$

How to define Dual Problem?

鏡射 $\phi(\vec{x}_i)$

$$\mathcal{L} = \frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i\left[y_i(\vec{w}^T\vec{x}_i+b)-1+\xi_i\right] - \sum_{i=1}^{N}\beta_i\xi_i$$

$$\frac{\partial}{\partial w}\mathcal{L} = w - \sum_{i=1}^{N}\alpha_i y_i \underline{x_i} = 0 \Rightarrow w = \sum_{i=1}^{N}\alpha_i y_i x_i \quad \text{_____ } \Phi$$

$\phi(\vec{x}_i)$ $\phi(\vec{x}_i)$

$$\frac{\partial}{\partial b}\mathcal{L} = -\sum_{i=1}^{N}\alpha_i y_i = 0 \Rightarrow \sum_{i=1}^{N}\alpha_i y_i = 0 \quad \text{——— ②}$$

$$\frac{\partial}{\partial \xi_i}\mathcal{L} = C - \alpha_i - \beta_i = 0 \Rightarrow \text{We know } \alpha_i, \beta_i \geq 0$$

$$\Rightarrow \beta_i = C - \alpha_i \geq 0 \Rightarrow C \geq \alpha_i \quad \text{——— ③}$$

代回原式 Problem

$$\mathcal{L} = \frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i\left[y_i(\vec{w}^T\underbrace{\vec{x}_i}_{\phi(\vec{x}_i)}+b) - 1 + \xi_i\right] - \sum_{i=1}^{N}\beta_i\xi_i$$

$$= \frac{1}{2}\sum_{j=1}^{N}\sum_{i=1}^{N}\alpha_i\alpha_j y_i y_j \phi^T(x_i)\phi(x_j) - \sum_{j=1}^{N}\sum_{i=1}^{N}\alpha_i\alpha_j y_i y_j \phi^T(x_i)\phi(x_j)$$

$$+\sum_{i=1}^{N}\alpha_i - \underbrace{\cancel{\sum_{i=1}^{N}\alpha_i y_i b}}_{0}$$

純量 and we know $\sum \alpha_i y_i = 0$

$$= \sum_{i=1}^{N}\alpha_i - \frac{1}{2}\sum_{j=1}^{N}\sum_{i=1}^{N}\alpha_i\alpha_j y_i y_j \phi^T(x_i)\phi(x_j)$$

Dual Problem Objective Function

Dual Problem:

$$\max \sum_{i=1}^{N}\alpha_i - \frac{1}{2}\sum_{j=1}^{N}\sum_{i=1}^{N}\alpha_i\alpha_j y_i y_j \phi^T(x_i)\phi(x_j)$$

$$\text{s.t. } \sum_{i=1}^{N}\alpha_i y_i = 0$$

$$0 \leq \alpha_i \leq C, \ \forall i = 1, \dots, N$$

KKT. condition

$\alpha_i[y_i(w^T\phi(x_i)+b)-1+\xi_i]=0, \forall i$

$\beta_i\xi_i=0, \forall i$

$\alpha_i, \beta_i \geq 0$

$\alpha_i+\beta_i=C$, $C>0$ 超参数

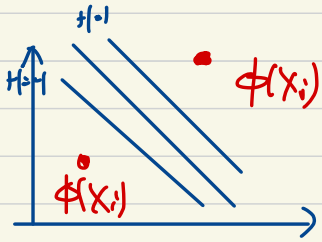
$\alpha \begin{cases} \alpha_i=0 \\ 0<\alpha_i<C \\ \alpha_i=C \end{cases}$

3种 case

Case 1°. $\alpha_i=0$

$\alpha_i=0$, $\because \alpha_i+\beta_i=C>0$

$\Rightarrow \beta_i>0 \Rightarrow \because \beta_i\xi_i=0 \Rightarrow \xi_i=0$



Conclusion: 在 feature map 上, 必在 margin 外, (也不在 H=1 H=-1 上)

Case 2°. $0<\alpha_i<C$

$0<\alpha_i<C \Rightarrow \because \alpha_i+\beta_i=C \Rightarrow \beta_i>0$

$\Rightarrow \beta_i\xi_i=0 \Rightarrow \xi_i=0$

由于 $\alpha_i>0$, 故, $y_i(w^T\phi(x_i)+b)-1+\xi_i=0$

if $y_i=1$, $w^T\phi(x_i)+b=1$

$\Rightarrow \phi(x_i)$ 在 $H=1$ 上

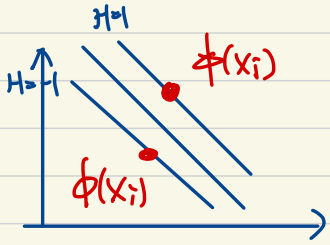
if $y_i=-1$, $w^T\phi(x_i)+b=-1$

$\Rightarrow \phi(x_i)$ 在 $H=-1$ 上

Case 2,

$\boxed{\text{可解 } b}$: $y_i(w^T\phi(x_i)+b^*)-1=0$

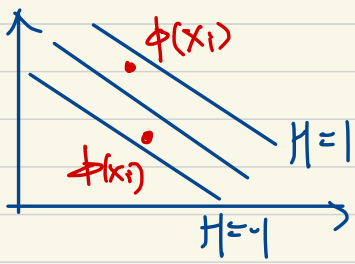
$$\Rightarrow y_i\left(\sum_{j=1}^{N}\alpha_j y_j \phi^T(x_j)\phi(x_i)+b^*\right)=1$$



$$b^*=\frac{1}{y_i}-\sum_{j=1}^{N}\alpha_j y_j \phi^T(x_j)\phi(x_i)$$

Case 3, $\alpha_i=C$

$\alpha_i=C$, $\alpha_i+\beta_i=C \Rightarrow \beta_i=0 \Rightarrow \because \beta_i\xi_i=0 \Rightarrow \xi_i>0$



$\because \alpha_i=C>0$

$$\Rightarrow y_i(w^T\phi(x_i)+b)-1+\underbrace{\xi_i}_{\neq 0}=0$$

若 $y=1$, $w^T\phi(x_i)+b=1-\xi_i$

若 $y=-1$, $w^T\phi(x_i)+b=-1+\xi_i$